

HealthConnect Week 4 Initial Analysis Document

AnalystLab Africa Experience Lab — Week 4 | Data Analytics Track

1. Project and Business Problem Understanding

HealthConnect Clinic is a fictional healthcare provider experiencing missed appointments, difficulty understanding no-show factors, inefficient use of appointment slots, repetitive patient enquiries and a need to improve patient engagement and administrative support.

Central project question: How can HealthConnect Clinic use data to reduce missed appointments and improve the patient support experience?

Data Analytics responsibility: understand the appointment data and identify how it can be used to investigate appointment attendance and no-show patterns.

2. Official Data Resources

Resource	Role
HealthConnect_Appointment_Data.csv	Primary appointment-level dataset for structure and quality review.
HealthConnect Data Dictionary	Official definitions, data types, examples and notes used to validate variable interpretation.

The Data Dictionary confirms that patient_id may recur across appointments, age represents adult patient age, age_group is derived for descriptive analysis, and appointment_outcome consists of Attended, No-Show and Cancelled.

3. Dataset Overview

The dataset contains 5,000 appointment records, 18 variables and 1,696 unique anonymised patients.

Area	Dictionary-confirmed meaning
Identifiers	appointment_id is the unique appointment identifier and primary key; patient_id is anonymised and may appear across multiple appointments.
Demographics	gender is recorded gender; age is patient age in years for adult patients; age_group is an age band derived from age.
Appointment details	appointment_type is the scheduled appointment type; appointment_day is derived from appointment_date; appointment_time is Morning, Afternoon or Evening.
Booking/history	booking_date and appointment_date define booking/scheduled dates; booking_lead_days is the days between them; previous_appointments and previous_no_shows are historical counts.
Reminders	reminder_sent indicates whether a reminder was sent; reminder_channel is the delivery channel and is None when no reminder was sent.

Access/operations	distance_to_clinic_km is approximate clinic distance; waiting_time_minutes is estimated waiting time.
Outcome	appointment_outcome is the final appointment status: Attended, No-Show or Cancelled.

4. Data Quality Assessment

There are 150 missing cells in the supplied 5,000 × 18 dataset. Missingness is concentrated in distance_to_clinic_km (90) and waiting_time_minutes (60).

Check	Result	Assessment
Fully duplicated rows	0	No exact duplicates detected.
Duplicate appointment_id	0	No duplicate primary keys detected.
Booking lead-day consistency	0	No mismatch between date difference and booking_lead_days.
Age/age_group consistency	0	No mismatch detected.
previous_no_shows > previous_appointments	0	No logical violation; matches dictionary constraint.
reminder_sent=No with channel populated	0	No contradiction detected.
reminder_sent=Yes with channel missing	0	No contradiction detected.
Missing distance	90	Limited missing values; treatment to be documented later.
Missing waiting time	60	Limited missing values; treatment to be documented later.

The missing reminder_channel values are not ordinary errors: the official Data Dictionary explicitly states that the channel is None when no reminder was sent. This state should therefore be preserved in later analysis rather than blindly imputed.

5. Variables Relevant to Attendance and No-Show Analysis

Variable	Why it matters
appointment_outcome	Primary outcome for attendance, no-show and cancellation analysis.
booking_lead_days	Tests whether the time between booking and appointment is associated with outcome.
previous_no_shows	Direct historical indicator for repeat non-attendance patterns.
previous_appointments	Provides appointment-history context.
reminder_sent	Supports reminder coverage and outcome comparisons.
reminder_channel	Supports comparison of delivery channels; None has a defined meaning when no reminder was sent.
distance_to_clinic_km	Supports investigation of access/travel patterns.
waiting_time_minutes	Supports investigation of operational experience and outcomes.

appointment_type	Allows outcome comparison across four appointment categories.
appointment_day	Allows day-of-week pattern analysis.
appointment_time	Allows Morning/Afternoon/Evening pattern analysis.
age / age_group	Supports descriptive demographic segmentation of adult patients.
gender	Supports descriptive segmentation; no causal interpretation should be assumed.

6. Relevant Business Questions

1. What proportion of scheduled appointments are Attended, No-Show and Cancelled?
2. Which appointment types have the highest no-show rates?
3. Is booking lead time associated with no-show rates?
4. Are patients with previous no-shows more likely to have another No-Show?
5. Is reminder coverage associated with appointment outcomes?
6. Among appointments where a reminder was sent, do outcomes differ by reminder channel?
7. Do no-show rates vary by appointment day or time period?
8. Is distance to the clinic associated with no-show patterns?
9. Does estimated waiting time show any relationship with appointment outcomes?
10. Do attendance outcomes differ across age groups or recorded gender categories?
11. How should cancellations be reported separately from no-shows so that the two outcomes are not conflated?

7. Potential KPIs for Subsequent Stages

Potential KPI	Linked business question	Why relevant
No-Show Rate	What proportion of scheduled appointments are missed?	Direct measure of the core missed-appointment problem.
Attendance Rate	What proportion of appointments are attended?	Measures successful completion and complements No-Show Rate.
Cancellation Rate	How much scheduled activity ends in cancellation?	Quantifies cancellations while keeping them distinct from no-shows.
Average waiting time	How long do patients wait?	Measures service efficiency.
Total Patients	How many unique patients are being served?	Show Overall patients volume and helps with resources planning.

8. Initial Analysis Approach

1. Preserve the original CSV and perform any transformations in a separate processed copy.
2. Use the official Data Dictionary as the authoritative reference for field interpretation.
3. Profile identifiers, dates, missingness, categories and numeric ranges.
4. Define Attended, No-Show and Cancelled as distinct outcomes.
5. Calculate the selected KPIs in the next analysis stage.
6. Compare no-show outcomes by appointment type, lead time, previous no-shows, reminder status/channel, day, time, age group, gender and distance.
7. Assess waiting time separately as an operational measure.
8. Use appointment dates for time-based patterns.

9. Treat descriptive relationships as associations rather than proof of causation.
10. Translate validated patterns into business insights and inputs for later project development.

9. Initial Dataset Observations

The supplied data contain 2,423 No-Show records (48.46%), 2,314 Attended records (46.28%) and 263 Cancelled records (5.26%). These are initial descriptive observations and are not presented as the final Week 4 KPI analysis.

The Data Dictionary confirms that reminder_channel = None is expected when no reminder was sent. It also confirms that patient IDs may repeat and previous_no_shows cannot exceed previous_appointments.

10. Assumptions, Limitations, Risks and Dependencies

Area	Consideration	Implication
Synthetic/anonymised data	Dataset uses fictional/synthetic and anonymised information.	Findings should not be presented as real clinical population evidence.
Missing distance/waiting time	Both contain limited missing values.	Document treatment before segment-level analysis.
Reminder channel	None is expected when no reminder is sent.	Do not blindly impute this field.
Repeated patients	patient_id may appear across multiple appointments.	Distinguish appointment-level from patient-level measures.
Outcome separation	Three final statuses exist.	Keep Cancelled separate from No-Show unless a later definition says otherwise.
Association vs causation	Descriptive differences do not prove causes.	Use cautious interpretation and appropriate methods later.
Reminder comparison	Reminder assignment may reflect operational choices.	Avoid claiming causal reminder effectiveness from simple comparisons.
Week 4 scope	Foundation stage only.	No final dashboard is required.

11. Proposed Week 5 Focus

Subject to the official Week 5 brief, move into validated analysis: prepare the analysis-ready data, calculate the agreed KPIs, investigate the strongest no-show patterns, identify actionable findings and begin developing decision-oriented outputs.